

$$\theta \leftarrow \theta + \eta \left(\frac{1}{n} \sum_{i=1}^n \log(p(\theta)) + \frac{1}{n} \sum_{i=1}^n \log(p(y_i | x_i, \theta)) \right) + \epsilon$$

Bayesian Deep Learning

UNIS: Activation fct's: $\tanh(x) = 2\sigma(2x) - 1$ (1 - $\sigma(2x)$) (saturated)

Initialization: $\sigma(\mathbf{I}) = \frac{\exp(\mathbf{I})}{\sum_j \exp(\mathbf{I}_j)}$ $\rightarrow \rho = \sigma(\mathbf{I})$ (saturated)

UNIS: min. MSE across all dev with Gaussian likelihood

Bayesian NNs

UNIS: $\mathbf{y} | \mathbf{x} \sim \mathcal{N}(\mathbf{y} | \mathbf{x}; \mathbf{W}, \sigma^2)$

UNIS: Instead of training $\mathbf{W}$, train $p(\mathbf{W} | \text{data})$ ^{weights of}

UNIS: **Probabilistic Noise:** $\mathbf{y} | \mathbf{x}, \mathbf{W}, \sigma^2 \sim \mathcal{N}(\mathbf{y} | \mathbf{x}; \mathbf{W}, \sigma^2)$

UNIS: $p(\mathbf{W}) = p(\mathbf{W}_0) \cdot \sigma^2(\mathbf{W}_0) \cdot \exp\left(-\frac{1}{2} \sum_{i,j} \frac{(\mathbf{W}_{ij} - \mathbf{W}_{ij}^0)^2}{\sigma^2(\mathbf{W}_{ij})}\right)$

UNIS: $p(\mathbf{y} | \mathbf{x}, \mathbf{W}, \sigma^2) = \text{const} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} \left(\frac{\mathbf{y} - \mathbf{W}\mathbf{x}}{\sigma^2}\right)^T \left(\frac{\mathbf{y} - \mathbf{W}\mathbf{x}}{\sigma^2}\right)\right)$

UNIS: can estimate $\mathbf{y}$ by accuracy or large σ^2 unc.

UNIS: It is penalized for large σ^2 - keeps stochastic

UNIS: $p(\mathbf{W})$ with prior $p(\mathbf{W}_0) \sim \mathcal{N}(\mathbf{W}_0, \sigma^2(\mathbf{W}_0))$

$$\begin{aligned} \arg \max_{\theta} & \log(p(\theta)) + \sum_{i=1}^n \log(p(y_i | x_i, \theta)) \\ \arg \min_{\theta} & \frac{1}{2n} \|\theta\|^2 + \frac{1}{2n} \sum_{i=1}^n (y_i - f(x_i; \theta))^2 \end{aligned} \quad \lambda = \frac{1}{6\sigma^2}$$
[illegible]
$$\text{mic Uncertainty} \approx \frac{1}{n} \sum \sigma^2(x^*; \theta) = \frac{1}{n-1} \sum_{i=1}^n (\bar{y}(x^*) - \hat{y}(x^*))^2$$

Chain Monte Carlo
 from posterior: $p(\gamma) \propto \prod_{i=1}^n p(\gamma_i | x_i, y_i)$
 sampling: keep track of a sequence of $\{\theta_1, \dots, \theta_n\}$. Use the weights for inference
 e.g. $\bar{\theta} = \frac{1}{n} \sum_{i=1}^n \theta_i$, $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \theta_i w_i$
 means: $\mu = \frac{1}{n} \sum_{i=1}^n \mu_i$, $\sigma^2 = \frac{1}{n} \sum_{i=1}^n \sigma_i^2$
 correct: set weight θ_i to zero. View as Vark
 $\theta_i = \frac{1}{n} \sum_{j=1}^n \theta_j$, $q(\theta_i | \gamma) = p(\theta_i | \gamma) \cdot \frac{1}{n} \sum_{j=1}^n \theta_j$
 original weights of N . Learn $\hat{\theta}$ by max ELO
 dev. learn not traceable for $\hat{\theta}$. Solution:
 $\theta_i(\gamma) = p(\theta_i | \gamma, 0, 1) - (1-p)(\theta_i | \gamma, 1, 1)$
 $\theta_i(\gamma) \approx \frac{1}{n} \sum_{j=1}^n \theta_j$ for large n
 e.g. $\theta_i(\gamma) = \frac{1}{n} \sum_{j=1}^n \theta_j$

$$A_{20} = \theta = 20^\circ + \epsilon, \quad 2(1) = 2000 \quad + \frac{P}{2} \parallel \underline{2(1)}_2 \quad + \epsilon_1$$

$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \sum_{i=1}^m \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta_i)} \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta_i)} \approx \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \sum_{i=1}^m \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta_i)} \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta_i)}$

average predictions of m neural networks

Stochastic Ensembles

weights of m NNs. Choose m train sets via
 (sampling or use random init in nets for variability)

in MAP estimates $\hat{\theta}_i = \arg \max_{\theta} \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta)}$

$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \sum_{i=1}^m \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta_i)} \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta_i)} \approx \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \sum_{i=1}^m \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta_i)} \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta_i)}$

Calibration

chance: maximise lower bound: $\frac{1}{m} \sum_{i=1}^m \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta_i)} \mathbb{E}_{\mathbf{y} \sim \mathcal{P}(\mathbf{y}|\mathbf{x}, \theta_i)}$

Bayesian Quantiles: Group predictions of val set

M bins of size $\frac{1}{M}$ according to $P(\hat{y}_i | x_i)$

[illegible]
$$r_{KS}: \log(N(\theta|m, \Sigma^{-1})) \propto -\frac{1}{2} \theta^T \Sigma \theta + m \cdot \theta \cdot 0 + \text{const}$$

- REINFORCE is not guaranteed to converge to J^*
 ↳ may get stuck in local optima.

[illegible]

Partial: Couaction: Noise filtering: Partial DE deterministic policy gradients, DDPG possibly non-action-value replay buffer $D = \{o, a, r, s\}$ pick action $a_t = \pi_{\theta}(s_t)$ & fire $a_t \in \mathcal{A}(s_t)$ observe r_t, s_{t+1} & replay buffer D sample (o, a, r, s) from D policy improvement step use iterations d for each sample (o, a, r, s) in D, compute the label $y = r + \gamma V(s_{t+1}) - V(s_t)$ actor update $\theta \leftarrow \theta + \alpha \nabla_{\theta} (Q(s_t, a_t) - V(s_t))$ critic update $w \leftarrow w + \alpha \nabla_w (Q(s_t, a_t) - y)$ end for $(1 - \gamma)V(s_t) \leftarrow y + \gamma V(s_t)$	Two delayed DDPG: 2 critic networks for grad of a_t w.r.t action a 2 training policies 2 stochastic policies
Constrained Policy: $\max_{\pi} \max_{\pi'} J(\pi, \pi')$ $J(\pi, \pi') = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$ $J(\pi, \pi') \leq \epsilon$ $\mathbb{E}_{(s, a) \sim \pi} [r(s, a)] \leq \epsilon$	Constrained Policy: maximize $J(\pi, \pi')$ subject to $J(\pi, \pi') \leq \epsilon$
Optimized Policy: $J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$ $J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$ $J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$	Optimized Policy: maximize $J(\pi)$ subject to $J(\pi) \leq \epsilon$
Objective: Policy $J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$ Gradient: $\nabla_{\theta} J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [\nabla_{\theta} r(s, a)]$ Loss: $L(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$ Maximum Entropy Policy: Policy Exploration: $J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$ Policy Exploration: $J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$	Objective: Policy $J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$ Gradient: $\nabla_{\theta} J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [\nabla_{\theta} r(s, a)]$ Loss: $L(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$ Maximum Entropy Policy: Policy Exploration: $J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$ Policy Exploration: $J(\pi) = \mathbb{E}_{(s, a) \sim \pi} [r(s, a)]$

HF: 1. Learn reward model r_θ from human feedback. 2) Use r_θ to determine T_{θ^*}

[illegible]